

The Editor-in-Chief
IEEE Transactions on Multimedia
IEEE Signal Processing Society

Dear Editor-in-Chief,

We respectfully submit our manuscript, “**NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction,**” for consideration as an original research article in *IEEE Transactions on Multimedia* (TMM).

Scope alignment. NARCIS addresses a core multimedia communications problem: how to transmit authenticated messages by selecting and sending unchanged natural images from a shared index, rather than by modifying a carrier. The technical pipeline is a multimedia processing contribution — a compact self-supervised encoder that maps images to channel-invariant descriptors, a quantile codebook that converts the descriptor space into balanced message symbols, and a multi-attack qualification stage that retains only images whose labels remain stable under channel distortions. These components sit squarely within TMM’s scope on multimedia content analysis, multimedia security, and image-based communications. The detectability evaluation (three detector classes, five partitions per dataset, calibration and holdout conditions) also directly addresses the multimedia security analysis that TMM regularly publishes.

Key technical contributions.

- **Feasible-codebook operating point (Contribution 1).** Existing coverless systems report nominal capacity or average symbol accuracy, neither of which verifies that a given message can actually be transmitted from a finite, qualified index. NARCIS introduces the *feasible operating point*: the largest symbol alphabet for which every attack-qualified bucket satisfies the multiplicity required by every protected coded message. Over 1,575 complete-message trials on BOSSBase 1.01 (grayscale) and Caltech-101 (RGB), NARCIS recovered 100% of authenticated plaintexts under 21 channel conditions, including eight unseen holdout strengths.
- **Self-supervised encoder and quantile codebook (Contribution 2).** A four-block convolutional network trained with a variance–invariance–covariance objective produces 64-dimensional normalised descriptors. A quantile partition of the first principal component yields balanced symbol classes. Attack qualification (12 calibration transforms) removes geometrically unstable images before indexing. A calibration-locked random projection search recovers 210/210 complete-message trials while increasing the minimum stable bucket from 92 to 161 (BOSSBase seed 11). Net throughput at a 16-symbol operating point scales from 0.184 to 1.213 bits per cover for 8- to 128-byte payloads.
- **Detectability and ablation evaluation (Contribution 3).** Selection bias is quantified with three detector classes (global logistic regression on dense residuals, residual-feature ExtraTrees, selection CNN) over five partitions each of BOSSBase and Caltech-101. BOSSBase AUCs include chance. Caltech-101 classical detectors identify a weak but reproducible selection shift (global logistic AUC 0.531, 95% CI [0.504, 0.557]; ExtraTrees 0.533, [0.504, 0.562]); 20 repeated CNN fits on the most sensitive partition confirm a partition-specific signal (mean AUC 0.533, 95% CI [0.517, 0.548]), reported transparently rather than dismissed. Removing attack qualification on BOSSBase seed 11 reduces complete-message success from 210/210 to 142/210 (67.6%), establishing qualification as the dominant robustness mechanism.

Novelty and positioning. Selection-based coverless systems have been evaluated at the descriptor level — average symbol accuracy, nominal capacity — but have not verified message-level feasibility over a finite qualified index, nor integrated payload authentication, replay control, and error correction into a single testable system. TMM has published related work in learned descriptors for coverless steganography (Ren–Wu 2024; Guo–Ping 2026) and in multimedia security evaluation. NARCIS complements this body of work by introducing the feasible-codebook criterion as the primary evaluation unit and providing systematic cross-dataset, multi-detector detectability evidence. The paper explicitly positions its measured capacity against prior reported values without overstating cross-protocol comparisons.

Author declarations.

- The manuscript is original, has not been published, and is not under consideration elsewhere. A previous version was desk-rejected by *Journal of Information Security and Applications* (Ms. No. JISAS-D-26-03622) for scope reasons and has since been substantially revised; this submission is not a transfer.
- All four authors approved the final manuscript and author order. WYSAWIS (cited in the related work) shares authors with this submission and is explicitly identified as such in the paper; it is used as published qualitative context, not as an experimental baseline.
- No competing interests and no external funding.
- No human subjects, personal data, or animals are involved. Implementation, fixed seeds, experimental results, and manuscript sources are available at github.com/EkodeckStephane/NARCIS.

We believe NARCIS will interest TMM readers working on multimedia steganography, learned image representations, and multi-

media security evaluation. We welcome the opportunity to provide any additional information the Associate Editor may require.

Yours sincerely,

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